SaaSafras Case Study — Maximizing Revenue
Objective:
To determine the optimal allocation of a 20-person team across three key functions—New Business Acquisition, Account Management, and Support—**over 24 months**. The goal is to **maximize cumulative revenue** and customer base while minimizing churn.

Strategy Development and Decision-Making Process
From pragmatic reasoning, effectively determining the optimal solution for maximizing revenue requires a brute-force optimization technique, as other methods aren't practical,
Using combinatorial mathematics (stars and bars method), the total number of strategies possible for this problem was determined to be 231^{24}, which is astronomically large and impractical to test manually.
By stars and bars, with n=20 team members and k=3 roles, the number of allocations

$$\binom{n+k-1}{k-1}$$

$$\binom{20+3-1}{3-1} = \binom{22}{2}$$

$$Number\ of\ allocations\ per\ decision\ unit = \frac{22\times21}{2\times1} = 231$$

Since the strategy can change every month, there are 24 decision points. Hence,

$$Total\ strategies = 231^{24} = 5.34 \times 10^{56}$$

At first glance, I came up with the allocation strategy below:

MODEL FALSE				
Month	Support	New Business Acquisition	Account Mgmt.	Reason
1-6	5	10	5	Prioritize rapid customer acquisition while maintaining support and initiating revenue growth
7-12	5	5	10	Shift focus to maximizing revenue growth from existing customers with balanced support.
13-24	10	5	5	Emphasize customer retention and satisfaction to stabilize long-term revenue.

Alert: This model is not the solution.

Using the formulas and constant recaps in the Parametric section below, MODEL FALSE yielded a cumulative revenue of $\$3,623,944$ — *See Matlab code SaaS_Model_False attached.*

Parametrics

Constants Recaps	
Initial Customers	1,000
Organic Growth	25 new customers per month
Baseline Churn Rate	10% (0.10)
Baseline Revenue per Customer:	$100
New Customers per Team Member (New Business)	5
Account Manager Capacity	25 customers per team member
CSAT Improvement (Support Team)	Reduces churn by 15% per member

Formulas
$Adjusted\ Churn\ Rate\ = Baseline\ Churn\ Rate\ 0.85^{Support\ Team\ Members}$
$Churned\ Cust.\ = Cust.\ Base\ Adjusted\ Churn\ Rate$
New Cust. from New Business =NB Team Members × New Cust. per Team Member
$Total\ New\ Cust. = Organic\ Growth + New\ Cust.\ from\ New\ Business$
$New\ Cust.\ Base = Prev.\ Cust.\ Base - Churned\ Cust. + Total\ New\ Cust.$
$Managed\ Cust. = Acc.\ Mgmt.\ Team\ Members \times Acc.\ Manager\ Capacity$
$Non - Managed\ Cust. = New\ Cust.\ Base - Managed\ Cust.$
$Rev.\ from\ Managed\ Cust. = Managed\ Cust. \times Baseline\ Revenue\ per\ Cust. \times 1.20$
$Rev.\ from\ Non\ Managed\ Cust. = Non\ Managed\ Cust. \times Baseline\ Revenue\ per\ Cust.$
$Total\ Revenue = Rev.\ from\ Managed\ Cust. + Rev.\ from\ Non - Managed\ Cust.$

Although reasonable, and highly profitable, MODEL FALSE cannot be deemed as the most optimal strategy for resource allocation. Hence a brute-force optimization code combined with parallel computing was leveraged to solve the problem — *via Matlab.*

Code explanation

Given the long simulation time, and computational resources required to determine the best, out of 231^{24} strategies, the 24-month planning horizon was divided into manageable 6-month intervals, significantly reducing the computational complexity by narrowing the search space at each step. This interval-based strategy simplifies the problem, allowing for a faster convergence to an optimal solution within a practical time limit — *See SaaSafras_Six_Code attached.*

OPTIMAL SOLUTION AND REASONING

Since the strategy can change every 6 months, there are 4 decision points. Hence, the total strategies tested $= 231^4 = 2.85$ billion strategies. After approximately 8hrs of run time, the optimal solution was determined to be:

OPTIMAL SOLUTION MODEL					
Month Interval	Support Team Size	New Business Team Size	Account Management Team Size	Cumulative Revenue ($)	Cumulative Customer Base
1-6	8	12	0	717,000	1327
7-12	10	10	0	1,613,200	1609
13-18	8	0	12	2,899,577.12	1507
19-24	2	0	18	**4,184,692.8**	1089

Note: This is the Optimal Solution

Explanation of Strategic Team Allocation Decisions

Months 1-6: Aggressive Customer Acquisition

- **Support (8 members):** Ensured new customers were satisfied and reduced churn, maintaining the momentum of growth.
- **New Business Acquisition (12 members)**: Maximizing customer acquisition was crucial at the start to build a strong base quickly. This set the foundation for future revenue growth
- **Account Management (0 members):** Not prioritized initially, as the focus was on expanding the customer base.

$Adjusted\ Churn\ Rate = 0.1 \times 0.85^8 = 0.0272 = 2.72\%$

The reasoning is centered around increasing the customer base and retaining both new and existing customers, which directly affects future revenue potential.

Months 7-12: Balancing Growth and Retention

- **Support (10 members)**: Increased focus on customer retention to maintain the growing base, crucial for sustaining long-term revenue.
- **New Business Acquisition (10 members)**: Continued customer acquisition while slightly reducing resources to allocate more to retention.
- **Account Management (0 members)**: Still deprioritized as building a larger customer base took precedence over immediate revenue from existing customers.

$$Adjusted\ Churn\ Rate = 0.1 \times 0.85^{10} = 0.0197 = 1.97\%$$

Again, Account Management was not introduced during this phase because the focus remained on maximizing the customer base and ensuring their satisfaction. The compound effect of customer retention at this stage was seen as more beneficial for long-term revenue than immediate revenue growth from existing customers.

Months 13-18: Shifting to Revenue Maximization

- **Support (8 members):** Sufficient to maintain customer satisfaction and minimize churn.
- **New Business Acquisition (0 members)**: No new customer acquisition; the customer base was large enough to focus on revenue.
- **Account Management (12 members):** Focused on increasing revenue from existing customers, leveraging the established base to maximize returns.

$$Adjusted\ Churn\ Rate = 0.1 \times 0.85^{8} = 0.0272 = 2.72\%$$

By the midpoint of the 24-month period, a substantial customer base had been established. Allocating all 12 available resources to Account Management was a strategic move to leverage the compounding effect of revenue growth. Each Account Manager was able to increase the revenue from their managed customers by 20% month-over-month, up to a cap of 6 months. This strategy was designed to maximize the per-customer revenue as quickly as possible, especially considering the churn rate and the stable customer base.

Months 19-24: Maximizing Revenue Potential

- **Support (2 members)**: Minimal support as the focus was on revenue, accepting a slight decrease in customer satisfaction.
- **New Business Acquisition (0 members)**: No new customers; the goal was to extract maximum revenue from the existing base.
- **Account Management (18 members)**: Maximized revenue extraction through intense focus on existing customers, aiming for the highest possible cumulative revenue.

$$Adjusted\ Churn\ Rate = 0.1 \times 0.85^{2} = 0.0723 = 7.23\%$$

In the final phase, the strategy was fully geared towards maximizing cumulative revenue. With 18 team members allocated to Account Management, the goal was to capitalize on the compounding revenue effects achieved in the previous months.
With customer acquisition and churn rates stabilized, the emphasis was on ensuring that the existing customer base continued to generate as much revenue as possible, even if it meant accepting a slight decline in customer satisfaction and base size.

Outcomes

- **Cumulative Revenue**: The strategy led to a cumulative revenue of $4,184,692.80.
- **Customer Base**: The customer base peaked at 1,609 in the 7-12 month interval but saw a reduction toward the end due to the strategic shift towards revenue maximization over customer acquisition.

Conclusion
The team allocations were driven by the need to first build a strong customer base and then shift to maximizing revenue from that base. Each decision was aimed at balancing growth, retention, and revenue optimization at the right times to achieve the highest possible cumulative revenue by the end of the 24 months

Bonus: Magic Wand
Recommended Variable Improvement: Customer Satisfaction (CSAT)
If given the opportunity to improve one variable going into year 3, I would focus on enhancing Customer Satisfaction (CSAT). This choice aims to address the root causes of churn, maximizing long-term revenue, and improving retention and stability.

Rationale for Improving

1. **Reduction in Churn Rate**: Improving CSAT directly lowers churn, subsequently improving customer retention. Lower churn stabilizes the customer base, reducing the need for aggressive new customer acquisition and fostering a more predictable revenue stream.
2. **Increase in Customer Lifetime Value (CLV)**: High CSAT leads to increased customer loyalty and higher CLV. Satisfied customers tend to stay longer and spend more, positively impacting overall revenue and supporting sustainable growth.
3. **Enhanced Referral and Organic Growth**: Satisfied customers are more likely to refer others, which generates additional revenue through organic growth. This reduces customer acquisition costs and supports a more cost-effective growth model.

Implementation Strategies

1. **Invest in enhanced support training and tools** to improve support team effectiveness, targeting at least a 2% point per support agent.

***From Case Study**: Each support agent increases CSAT by 1 percentage point. Each point of CSAT leads to a 15% relative decrease in churn, with a compounding effect (the next CSAT point decreases 15% of 85%, etc.).*

$Churn\ Rate\ = Baseline\ Churn\ Rate\ \times\ 0.85^{Support\ Team\ Members}$

If we achieve a 2 percentage point per support agent, the new church rate then becomes

$New\ Churn\ Rate\ = Baseline\ Churn\ Rate\ \times 0.85^{2\times Support\ Team\ Members}$

Previously, one support agent will reduce the churn rate from 10% to $0.1 \times 0.85 = 8.5\%$, and 2 support agents will reduce it to $0.1 \times 0.85^2 = 7.2\%$

With the *New Churn Rate*, one support agent will reduce the churn rate from 10% to $0.1 \times 0.85^2 = 7.2\%$, and 2 support agents will reduce it to $0.1 \times 0.85^{2\times2} = 5.2\%$

2. **Implement customer feedback systems** to identify and address the issue. This will significantly improve the quality of the product and or service.
   As you fix issues and make improvements based on feedback, overall customer satisfaction goes up, which can lead to higher customer loyalty and positive word-of-mouth — which means, more organic growth

By improving CSAT, you turn the feedback into action. As the saying goes, "*The way you get them is the way you lose them*." Satisfied customers stay loyal, while unhappy ones may leave.

Appendix

Detailed Monthly Allocation and Financial Performance Data

Month	SupportTeamSize	NewBusinessTeamSize	AccountManagementTeamSize	CumulativeRevenue	CumulativeCustomerBase
1	8	12	0	105800	1058
2	8	12	0	217300	1115
3	8	12	0	334300	1170
4	8	12	0	456700	1224
5	8	12	0	584300	1276
6	8	12	0	717000	1327
7	10	10	0	854600	1376
8	10	10	0	997000	1424
9	10	10	0	1144100	1471
10	10	10	0	1295900	1518
11	10	10	0	1452300	1564
12	10	10	0	1613200	1609
13	8	0	12	1831879.52	1591
14	8	0	12	2048759.04	1573
15	8	0	12	2263938.56	1556
16	8	0	12	2477418.08	1539
17	8	0	12	2689297.6	1523
18	8	0	12	2899577.12	1507
19	2	0	18	3131346.4	1424
20	2	0	18	3355415.68	1347
21	2	0	18	3572284.96	1275
22	2	0	18	3782454.24	1208
23	2	0	18	3986423.52	1146
24	2	0	18	4184692.8	1089

Code1:

SaaS_Model_False.m

```
% Strategy Allocation Based on the Given Intervals
% Columns: SupportTeamSize, NewBusinessTeamSize,
AccountManagementTeamSize
%Strategy based on user input
strategy = [
   5, 10, 5;   % Months 1-6: Support = 5, New Business = 10, Account
Management = 5
   5, 5, 10;   % Months 7-12: Support = 5, New Business = 5, Account
Management = 10
   10, 5, 5;   % Months 13-18: Support = 10, New Business = 5, Account
Management = 5
   10, 5, 5;   % Months 19-24: Support = 10, New Business = 5, Account
Management = 5
];
% Constants
months = 24;
initial_customers = 1000;
organic_growth = 25;
baseline_churn = 0.1;
baseline_revenue_per_customer = 100;
new_customers_per_team_member = 5;
account_manager_capacity = 25;
```

```
% Initialize the customer base and total revenue
customer_base = initial_customers;
total_revenue = 0;
% Iterate through the 24 months, applying the strategy
for month = 1:months
   % Determine the strategy interval (1-6, 7-12, 13-18, 19-24)
   interval = ceil(month / 6);

   % Extract the current interval strategy
   support_team_size = strategy(interval, 1);
   new_business_team_size = strategy(interval, 2);
   account_management_team_size = strategy(interval, 3);

   % Calculate the max customers under management
   max_customers_under_management = account_management_team_size *
account_manager_capacity;

   % Adjust churn rate based on CSAT improvement from the support team
   churn_rate = baseline_churn * (0.85 ^ support_team_size);
   churned_customers = floor(customer_base * churn_rate);

   % Calculate new customers acquired
   new_customers = organic_growth + new_business_team_size *
new_customers_per_team_member;

   % Update the customer base
   customer_base = customer_base - churned_customers + new_customers;

   % Calculate revenue from managed and non-managed customers
   managed_customers = min(customer_base,
max_customers_under_management);
   non_managed_customers = customer_base - managed_customers;
   revenue_from_managed = managed_customers *
baseline_revenue_per_customer * (1 + 0.20)^min(month, 6);
   revenue_from_non_managed = non_managed_customers *
baseline_revenue_per_customer;
   total_revenue = total_revenue + revenue_from_managed +
revenue_from_non_managed;
end
% Display the total revenue and final customer base
fprintf('Total Revenue after 24 months: $%.2f\n', total_revenue);
fprintf('Final Customer Base after 24 months: %d customers\n',
customer_base);
```

Code 2:

SaaSafras_Six_Code.m

```
% Main Script
```

```
% Constants
months = 24;
initial_customers = 1000;
organic_growth = 25;
baseline_churn = 0.10;
baseline_revenue_per_customer = 100;
team_size = 20;
new_customers_per_team_member = 5;
account_manager_capacity = 25;
max_time = 86400; % Maximum time in seconds (24hrs)
% Initialize strategy and best values
initial_strategy = zeros(4, 5); % Modified for 4 intervals (6-month
strategies)
% Define the total number of strategies to explore in parallel
num_strategies = (team_size + 1) * (team_size + 1);
% Preallocate arrays to store the best results from each parallel worker
best_revenues = zeros(num_strategies, 1);
best_strategies = cell(num_strategies, 1);
% Terminate any existing parallel pool
if ~isempty(gcp('nocreate'))
   delete(gcp('nocreate'));
end
% Create a new parallel pool
parpool;
% Start the global timer
global_start_time = tic;
% Use parallel computing to explore strategies
parfor idx = 1:num_strategies
   % Check if global time limit has been reached
   if toc(global_start_time) > max_time
       % Terminate if time limit exceeded
       best_revenues(idx) = -Inf;
       best_strategies{idx} = [];
       continue;
   end

   [support_team_size, new_business_team_size] = ind2sub([team_size + 1,
team_size + 1], idx);

   if support_team_size + new_business_team_size <= team_size
       account_management_team_size = team_size - support_team_size -
new_business_team_size;
       strategy = initial_strategy;
       strategy(1, :) = [support_team_size, new_business_team_size,
account_management_team_size, 0, 0]; % Modified for 5 columns

       % Initialize the best revenue and strategy for this worker
       local_best_revenue = -Inf;
       local_best_strategy = strategy;
```

```
        % Start timing for this worker
        worker_tic = tic;
        % Explore remaining intervals sequentially for this starting
strategy
        [local_best_revenue, local_best_strategy] =
explore_strategies_parallel(2, strategy, local_best_revenue,
local_best_strategy, 4, team_size, initial_customers, organic_growth,
baseline_churn, baseline_revenue_per_customer,
new_customers_per_team_member, account_manager_capacity, max_time,
worker_tic);

        % Store the results
        best_revenues(idx) = local_best_revenue;
        best_strategies{idx} = local_best_strategy;
    end
end
% Gather the best overall strategy
[best_revenue, best_idx] = max(best_revenues);
best_strategy = best_strategies{best_idx};
% Initialize a table for the 6-month interval strategy
interval_strategy = zeros(4, 5); % Columns: SupportTeamSize,
NewBusinessTeamSize, AccountManagementTeamSize, Cumulative Revenue,
Cumulative Customer Base
% Calculate cumulative values every 6 months and output interval table
for interval = 1:4
    [total_revenue, final_customers, ~] =
simulate_dynamic_strategy(best_strategy(1:interval, 1:3), interval * 6,
initial_customers, organic_growth, baseline_churn,
baseline_revenue_per_customer, new_customers_per_team_member,
account_manager_capacity);

    % Fill in the strategy allocations
    interval_strategy(interval, 1:3) = best_strategy(interval, 1:3);
    interval_strategy(interval, 4) = total_revenue; % Cumulative revenue
    interval_strategy(interval, 5) = final_customers; % Cumulative
customer base
end
% Create the month-interval labels
interval_labels = strcat(num2str((1:6:19)'), '-', num2str((6:6:24)'));
% Create a table with the interval labels and strategy data
interval_table = table(interval_labels, ...
                       best_strategy(:, 1), ...
                       best_strategy(:, 2), ...
                       best_strategy(:, 3), ...
                       interval_strategy(:, 4), ...
                       interval_strategy(:, 5), ...
```

```
                    'VariableNames', {'MonthInterval',
'SupportTeamSize', 'NewBusinessTeamSize', 'AccountManagementTeamSize',
'CumulativeRevenue', 'CumulativeCustomerBase'});
% Output the interval results table
fprintf('Interval Cumulative Revenue and Customer Base:\n');
disp(interval_table);
% Save interval table as an Excel file
filename = 'Best_Strategy.xlsx';
writetable(interval_table, filename);
% Initialize a table to store monthly cumulative revenue and customer
base
monthly_strategy = zeros(months, 6); % Columns: Month, SupportTeamSize,
NewBusinessTeamSize, AccountManagementTeamSize, Cumulative Revenue,
Cumulative Customer Base
% Calculate cumulative values every month and output monthly table
for month = 1:months
   interval = ceil(month / 6);
   [total_revenue, final_customers, ~] =
simulate_dynamic_strategy(best_strategy(1:interval, 1:3), month,
initial_customers, organic_growth, baseline_churn,
baseline_revenue_per_customer, new_customers_per_team_member,
account_manager_capacity);

   % Fill in the month and strategy allocations based on the current
interval
   monthly_strategy(month, 1) = month;
   monthly_strategy(month, 2:4) = best_strategy(interval, 1:3);
   monthly_strategy(month, 5) = total_revenue; % Cumulative revenue for
the month
   monthly_strategy(month, 6) = final_customers; % Cumulative customer
base for the month
end
% Output the monthly results table
monthly_table = array2table(monthly_strategy, 'VariableNames', {'Month',
'SupportTeamSize', 'NewBusinessTeamSize', 'AccountManagementTeamSize',
'CumulativeRevenue', 'CumulativeCustomerBase'});
fprintf('Monthly Cumulative Revenue and Customer Base:\n');
disp(monthly_table);
% Save monthly table as an Excel file
monthly_filename = 'Monthly_Strategy.xlsx';
writetable(monthly_table, monthly_filename);
% Plot revenue growth over time (monthly)
figure;
plot(1:months, monthly_strategy(:, 5));
title('Revenue Growth Over Time (Monthly)');
xlabel('Month');
ylabel('Cumulative Revenue');
grid on;
% Plot customer base growth over time (monthly)
```

```
figure;
plot(1:months, monthly_strategy(:, 6));
title('Customer Base Growth Over Time (Monthly)');
xlabel('Month');
ylabel('Cumulative Customer Base');
grid on;
% Shut down parallel pool
delete(gcp);
% Function Definitions
% (The function definitions remain unchanged)
% Function Definitions
% Function to calculate the total revenue, final customer base, and churn
rate
function [total_revenue, final_customers, final_churn_rate] =
simulate_dynamic_strategy(allocations, months, initial_customers,
organic_growth, baseline_churn, baseline_revenue_per_customer,
new_customers_per_team_member, account_manager_capacity)
   customer_base = initial_customers;
   total_revenue = 0;
   for month = 1:months
       support_team_size = allocations(ceil(month/6), 1);
       new_business_team_size = allocations(ceil(month/6), 2);
       account_management_team_size = allocations(ceil(month/6), 3);

       max_customers_under_management = account_management_team_size *
account_manager_capacity;

       % Adjust churn rate based on CSAT improvement from the support
team
       churn_rate = baseline_churn * (0.85 ^ support_team_size);
       churned_customers = floor(customer_base * churn_rate);
       new_customers = organic_growth + new_business_team_size *
new_customers_per_team_member;

       % Update customer base
       customer_base = customer_base - churned_customers + new_customers;

       % Calculate revenue from managed and non-managed customers
       managed_customers = min(customer_base,
max_customers_under_management);
       non_managed_customers = customer_base - managed_customers;
       revenue_from_managed = managed_customers *
baseline_revenue_per_customer * (1 + 0.20)^min(month, 6);
       revenue_from_non_managed = non_managed_customers *
baseline_revenue_per_customer;
       total_revenue = total_revenue + revenue_from_managed +
revenue_from_non_managed;
   end
   final_customers = customer_base;
```

```
    final_churn_rate = churn_rate;
end
% Recursive function to explore all strategies (adjusted for parallel
execution)
function [best_revenue, best_strategy] =
explore_strategies_parallel(interval, strategy, best_revenue,
best_strategy, total_intervals, team_size, initial_customers,
organic_growth, baseline_churn, baseline_revenue_per_customer,
new_customers_per_team_member, account_manager_capacity, max_time,
worker_tic)
    % Check elapsed time
    if toc(worker_tic) > max_time
        return;
    end
    if interval > total_intervals
        % End of recursion, calculate the performance of the strategy
        [total_revenue, ~, ~] = simulate_dynamic_strategy(strategy,
total_intervals * 6, initial_customers, organic_growth, baseline_churn,
baseline_revenue_per_customer, new_customers_per_team_member,
account_manager_capacity);
        if total_revenue > best_revenue
            best_revenue = total_revenue;
            best_strategy = strategy;
        end
    else
        % Loop through all possible team allocations for this interval
        for support_team_size = 0:team_size
            for new_business_team_size = 0:(team_size - support_team_size)
                account_management_team_size = team_size -
support_team_size - new_business_team_size;
                strategy(interval, :) = [support_team_size,
new_business_team_size, account_management_team_size, 0, 0]; % Modified
for 5 columns
                % Recursively explore the next interval
                [best_revenue, best_strategy] =
explore_strategies_parallel(interval + 1, strategy, best_revenue,
best_strategy, total_intervals, team_size, initial_customers,
organic_growth, baseline_churn, baseline_revenue_per_customer,
new_customers_per_team_member, account_manager_capacity, max_time,
worker_tic);
            end
        end
    end
end
```